



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

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Switching Techniques

- In large networks, there are often multiple possible paths from a sender to a receiver.
- **Switching is a technique used to transfer data from one node to another through intermediate network devices.**
- **Switching techniques determine how communication paths/connections are established, how network resources are allocated, and how data is forwarded through the network.**
- Routing determines an appropriate path for data transmission when multiple paths exist, while switching/forwarding moves the data through the selected path.
- Switching techniques can support communication between one-to-one and one-to-many/many-to-many systems, depending on the type of network and switching method.

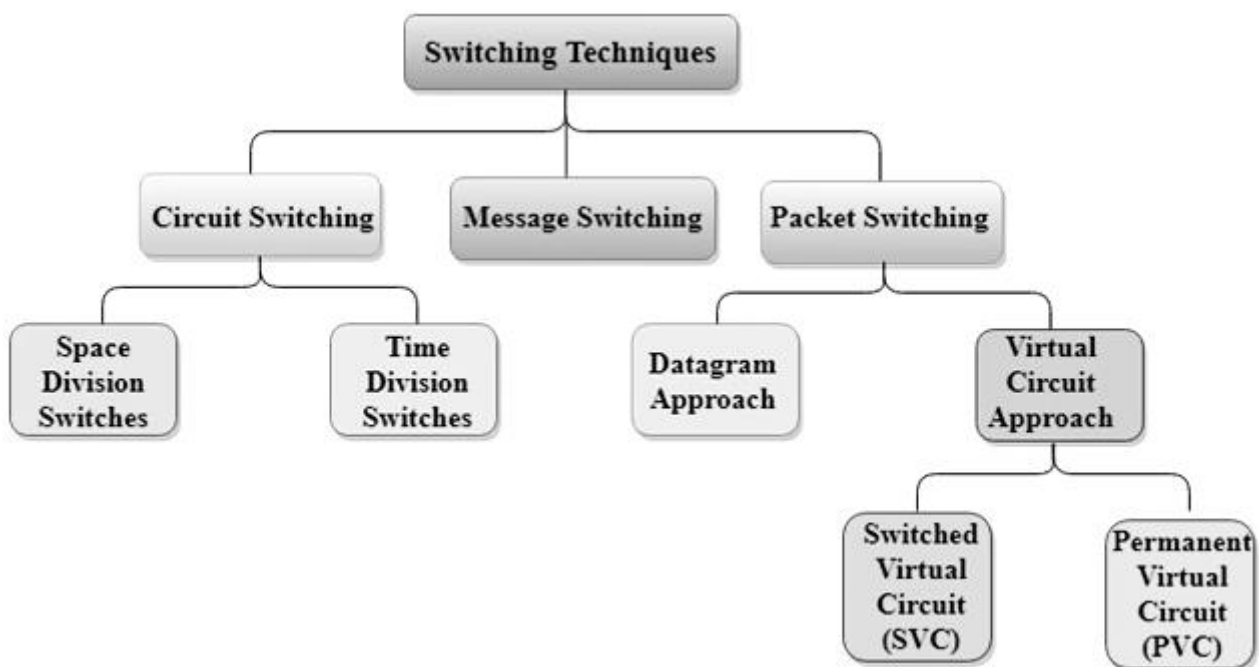
❖ **There are three typical switching techniques**

1. Circuit Switching

2. Message Switching

3. Packet Switching

Hierarchy switching



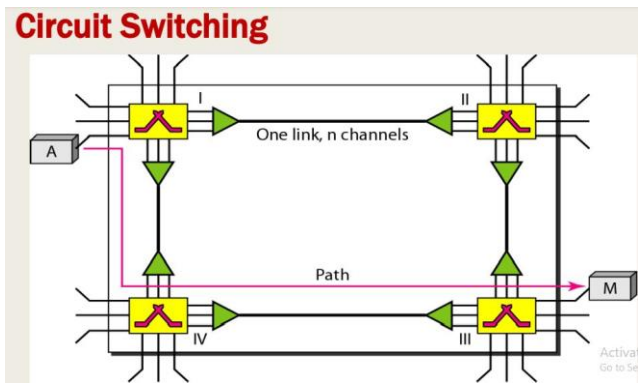
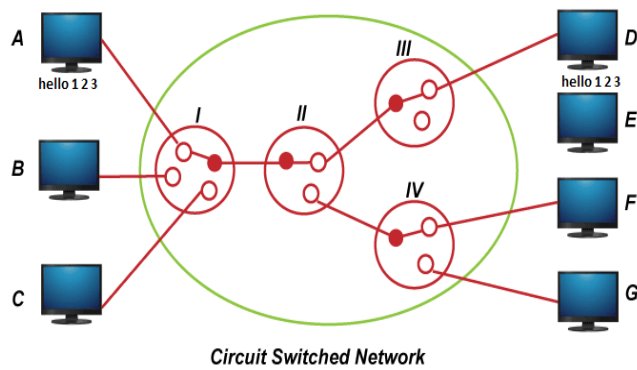
1. Circuit Switching

- A circuit-switched network is one of the simplest data communication methods in which a dedicated path is established/set-up(reserve) between the sending and receiving device.
- The physical links are interconnected through a series of switches, and the switches establish the required circuit.
- In circuit switching data flow continuously, no required to make data segment, packet, frame.
- In circuit switching no require any type headers/address.
- In circuit switching generally provides predictable delay and fixed bandwidth after the circuit has been established, but it can have an initial connection/setup delay.
- e.g – Rail line dedicated path, while road not dedicated path.

Note : Circuit switching was introduced in the late 19th century, around the 1876s, with the development of the telephone network by alexander graham bell.

Circuit switching has 3 main phase :

- Setup/Establishment phase** : Establish dedicated path sender to receiver
- Data transfer phase** : Send data/voice same dedicated path
- Teardown/termination phase** : Release circuit/path after communication



➤ Advantage

- In the case of Circuit Switching technique, the communication channel is dedicated.
- It has fixed bandwidth.

➤ Disadvantage

- Possible long wait to establish a connection, (10 seconds, more on long- distance or international calls.) during which no data can be transmitted.
- More expensive than any other switching techniques, because a dedicated path is required for each connection.
- Inefficient use of the communication channel, because the channel is not used when the connected systems are not using it.
- Billing is generally based on the duration of the connection such as per minute.

➤ Application/Use

- Telephone Network (use voice call)

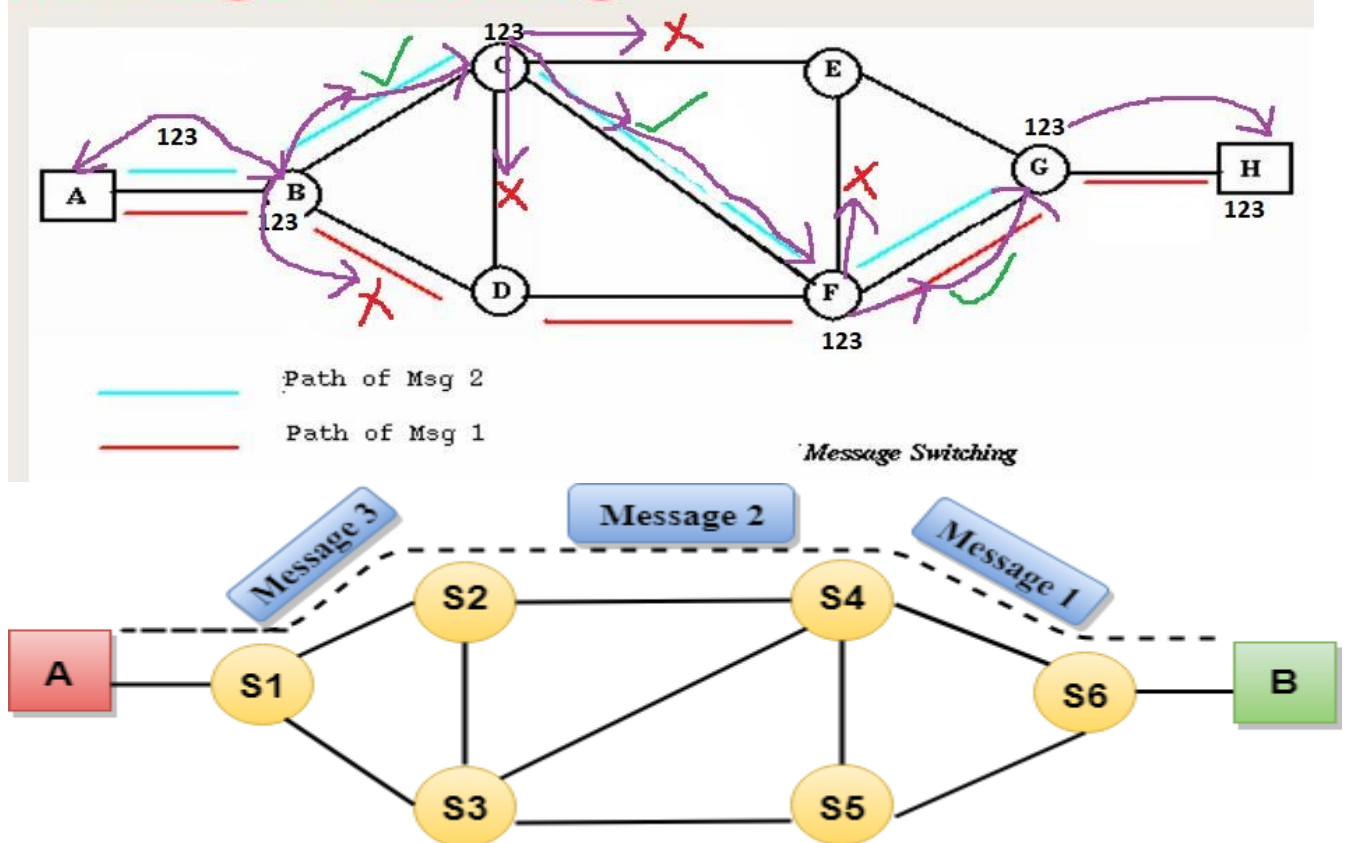
2. Message Switching (1960)

- In message switching there is no need to establish a dedicated path between two stations.
- Message switching send message use any path whether it is near or far.
- When a station sends a message, the destination address is added to the message.
- The message is then transmitted through the network, Countinuosly

in its entirety, from node to node.

- Each node receives the entire(पूरा) message, stores it in its entirety on disk, and then transmits the message to the next node. (e.g. when mobile switch off)
- This type of network is called a **store-and-forward network the entire message**. Storing device use hard disk, now using RAM.

Message Switching



➤ Advantages

- Channel efficiency can be greater compared to circuit switched systems, because more devices are sharing the channel.
- Traffic congestion can be reduced, because messages may be temporarily stored in route.
- Message priorities can be established due to store-and-forward technique.
- Message broadcasting can be achieved with the use of broadcast address appended in the message.

➤ Disadvantages

- Message switching is not compatible with interactive applications.
- Store-and-forward devices are expensive, because they must have large disks to hold potentially long messages.

➤ Application/use

- Mobile message

3. Packet Switching

- In the Packet switching, the message is broken down into smaller bits known as packets. Each packet contains a header which includes the source address, destination address, sequence number and control information. Each packet is stored briefly at intermediate nodes and then forwarded **store and forward** individual packets.
- If any packet is missing or corrupted, the message will be resent. If the packets are received in the right order, the acknowledgment message is delivered.

There are two methods to achieve Packet Switching

(i) Datagram Packet switching

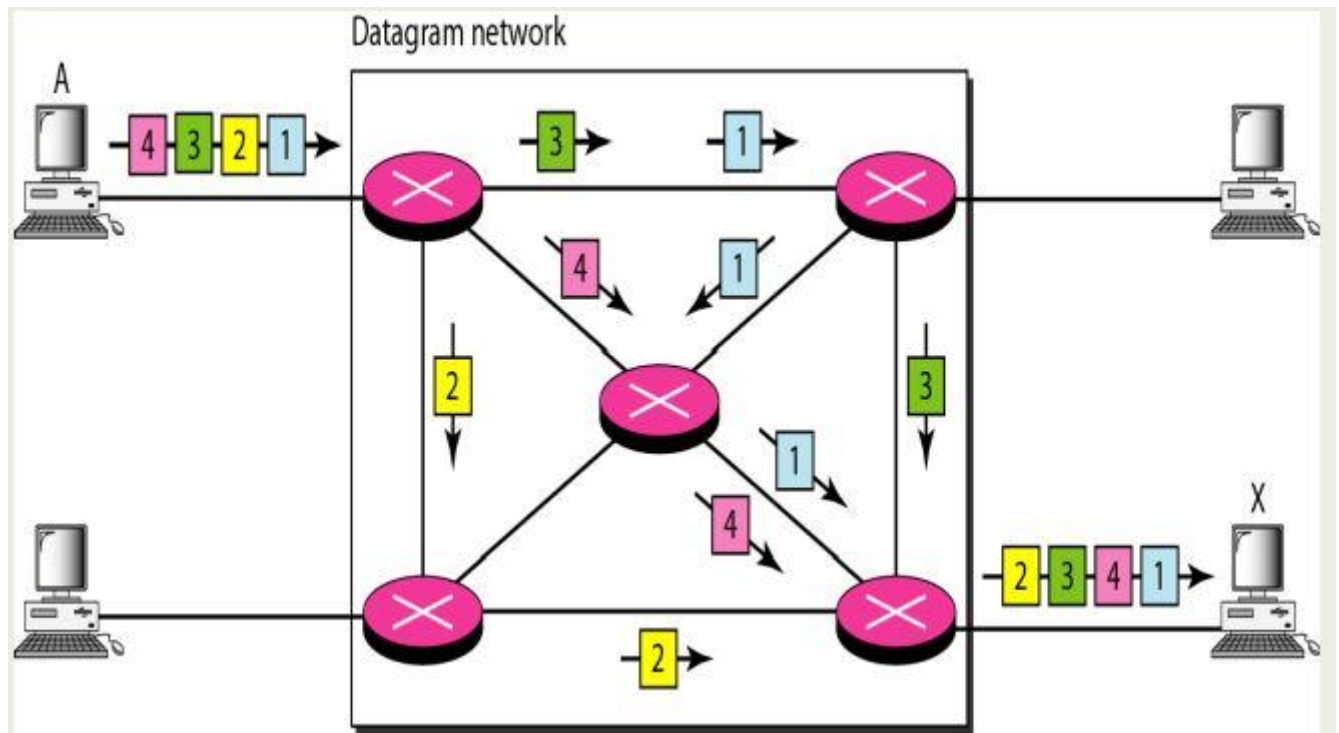
- Connectionless packet switching, no dedicated path is established before sending data.
- Each packet is treated independently
- Packets may arrive out of order and destination reassembles them into the original data.

(ii) Virtual circuit packet switching

- Connection-oriented packet switching
- A logical path (virtual circuit) is established before data transmission.
- Packet usually arrive in order.

Difference between Datagram and Virtual Circuit Packet Switching:

Datagram Packet Switching	Virtual Circuit Packet Switching
To forward packets, the node makes routing decisions.	There is no routing decision made by the nodes.
Congestion cannot develop since all packets go in opposite directions.	Congestion occurs when a node is overburdened and does not allow other packets to flow through.
It is more adaptable since each packet is considered as an individual entity.	It is not particularly adaptable.



➤ Advantage

- The advantage of packet switching is that many network users can share the same channel at the same time.
- Packet can be re-routed if there is any problem, such as, busy or disabled links.
- Packet switching is cost effective compare to message switching, because switching devices do not need massive (बड़े पैमाने पर) amount of secondary storage.

➤ Disadvantage

- Protocols for packet switching are typically more complex.
- It can add some initial costs in implementation.
- If packet is lost, sender needs to retransmit the data.
- Another disadvantage is that packet-switched systems still can't deliver the same quality as dedicated circuits in applications requiring very little delay - like voice conversations or moving images.
- e.g sender send message pincode 803116, and technical delay then receiver received message pincode 801613.

➤ Application/use

- Mobile message (now Email, whatsapp message)

Difference

Comparison between packet switching networks and circuit switching networks

Circuit Switching

- Dedicated transmission path
- Continuous transmission
- No storage of data/messages
- Same path for duration of connection
- Call setup required
- Overload may block call
- Busy signal if callee busy
- Fixed bandwidth transmission
- Charging (Money) by minute
- Fast enough for Interactive

Packet Switching

- No dedicated path
- Transmission of packets
- Packets may be stored
- Each packet may take different route
- No call setup needed
- overload increases packet delay
- no busy signal
- dynamic use of bandwidth
- Charging (Money) by packet
- Fast enough for Interactive (?)

What is switching techniques ? Discuss advantages of Packet Switching.

2021

10. Write short notes on **any four** of the following :

- (a) f print f ()
- (b) Recursion function
- (c) Call by reference function
- (d) Union
- (e) melloc()
- (f) Packet Switching

2023

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